Is correlation meaningfully correlated with causation? Crud, $1/f$ spectra, and an empirical null for small-effect observational claims

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Causal claims in observational research typically interpret an adjusted correlation as evidence of a causal effect. For this logic to work, adjusted correlations must be informative about underlying causal effects. We test this assumption using nine large datasets spanning health, neuroscience, psychology, genomics, politics, and vision. Across all nine, most pairwise correlations are classically significant; the residual crud scale σ_K shrinks only slowly under generic adjustment because shared variance is distributed across many components in approximately $1/f$ eigenvalue spectra. Even after removing $K = 10$ principal components, $r = 0.10$ does not exceed the conventional 95th-percentile magnitude threshold in six of the nine datasets, regardless of sample size: a magnitude unremarkable against same-pipeline background pairs. We propose the crud-aware diagnostic—a percentile rank in the empirical version, a z-test in a parametric small-study version—and use it to show that the typical published effect magnitude in observational medicine, nutritional epidemiology, common-variant genetics, and BWAS-style neuroscience does not clear the same-domain background. A decision-theoretic benchmark (Bayes error under a stylized Gaussian mixture) formalizes the magnitude-only limit. The diagnostic is a magnitude check complementary to identification: small adjusted correlations supported by RCT replication or mechanism (e.g., aspirin and cardiovascular mortality) can still encode real causation; magnitudes typical of randomly drawn pairs in the same domain cannot, by themselves.

causal inference | crud factor | background dependence | $1/f$ spectra | empirical null

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1. Introduction

Across biomedicine, social science, and behavioral research, causal claims increasingly rely on observational associations in large datasets. Randomized experiments remain the benchmark for counterfactual effects, but intervention is often infeasible, so small adjusted associations are routinely interpreted as evidence about causal mechanism. For this logic to work, adjusted correlations must be informative about their underlying causal effects. We test that assumption by asking how informative those associations remain in the presence of the background dependence structure typical of modern datasets. We focus on small-effect causal claims that depend on adjusted associations standing out from background dependence; large effects easily clear any reasonable background, and prediction tasks do not require causal interpretation.

Related work emphasizes that widespread weak dependence is common: Meehl's "crud factor" (1) argued that in large behavioral datasets, most pairs of variables are non-trivially correlated; **(author?)** (2) reviews the "everything is correlated" perspective. Sensitivity analyses (3, 4) ask the hypothetical question of how strong unobserved confounding would need to be to nullify a result; we ask the empirical question of what background dependence actually looks like in a domain. Approximate $1/f$ spectral structure has been documented across many complex systems (5). Our contribution connects these threads: we quantify the background correlation distribution across domains, link its slow shrinkage under generic adjustment to eigenvalue spectra, and propose a crud-aware test for interpreting small adjusted associations. A complementary literature documents that small published effects are often inflated (6–8); SI Appendix 6 surveys this evidence field by field and shows that the typical published effect would not clear our test in observational medicine, nutrition, common-variant genetics, or BWAS-style neuroscience.

A concrete example: a recent study reported a nominally significant association between folate intake and mortality in adults with type-2 diabetes ($r \approx 0.02$, $p_{\text{classical}} < 0.05$, $n \approx 8{,}000$) (9). Compared to the background distribution of correlations among NHANES variables, however, it falls in the middle: its crud-aware percentile rank is 0.19 (i.e., its magnitude is exceeded by 81% of same-pipeline reference pairs; Section H). The classical test detects that the correlation is nonzero; the percentile rank shows it is not distinctive against the background.

We operationalize the background dependence scale, for this paper, as the residual correlation standard deviation across uniformly sampled feature pairs after a fixed adjustment (regression on the top K principal components), and use it as the benchmark for whether an adjusted association is surprising. Whether this matches a domain's conceptual notion of background depends on whether the adjustment captures the domain's confounding structure. The empirical null is not a "no direct edge" null—in real domains, pervasive confounding ensures that uniformly sampled pairs share indirect dependence—it is the distribution of typical same-pipeline residual correlations. The argument is correspondingly weaker: a target sitting in the bulk cannot be cleanly read as evidence of a direct causal relation, because the same residual magnitude can arise from typical residual confounding alone. Across our nine domains, most small adjusted associations do not separate from the empirical null. We (i) document this empirically, (ii) trace it to approximately power-law eigenvalue spectra, (iii) prove a Bayes-error benchmark (Theorems 2–3 in SI Appendix 2) for magnitude-only classification under a stylized mixture, and (iv) propose a crud-aware test against the empirical null.

2. Results

A. Datasets. We analyze nine large datasets spanning neuroscience, genomics, psychology, political science, vision, and population health (Figure 1; Methods; SI Appendix 1 gives full descriptions and provenance): NHANES, precinct-level 2016 voting and Census demographics, GTEx v8 RNA-seq, Allen Institute mouse brain RNA-seq, HEXACO, CIFAR-10, Kay fMRI, Haxby fMRI, and Stringer mouse V1. The nine span dimensionalities (p from 83 to 6,000), raw sample sizes (n from ∼8,000 to ∼30,000; analysis subsamples to $n = 10{,}000$ where larger, see Methods), and measurement types (continuous neural signals, binary/ordinal survey items, count-based expression, pixel intensities), so patterns common to all cannot easily be attributed to a single data modality. This is a convenience sample of large public datasets; generalization beyond the nine remains an empirical question.

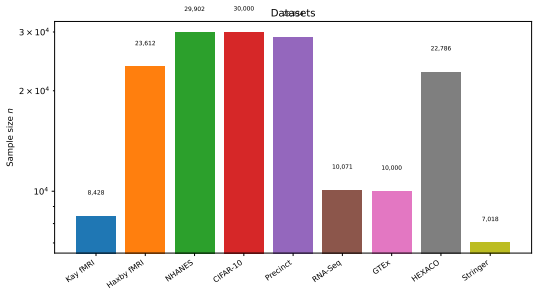

Fig. 1. Nine large public datasets span a wide range of sample sizes, dimensionalities, and measurement modalities. Bars show analysis sample size n per dataset on a log scale; see Table 1 for p and SI Appendix 1 for provenance.

B. Empirical finding: broad correlations persist after adjustment. For uniformly sampled feature pairs, we observe a broad spread of correlations (Figure 2, left; Table 1). Given the large sample sizes, classical iid correlation tests label most of these correlations as nominally significant, even though most random pairs are implausible candidates for direct causal relations. After PC regression at $K = 10$ (Methods; sensitivity across K in Table 1), the residual correlation distribution remains wide (Figure 2, right). The core empirical fact: the background correlation distribution is wide even after generic adjustment, so "significant" does not mean "distinctive." This is not a multiple-testing point: even ignoring p-values, the typical residual correlation scale after generic adjustment is substantial, so small adjusted associations are not distinctive evidence in the first place. Across the low-rank adjustments we tested (PC regression and Lasso, SI Appendix D), none collapsed the residual distribution on any of the nine datasets.

To give these numbers a concrete interpretation, for $r = 0.10$ to clear the conventional 95th-percentile crud-aware threshold ($z_{\mathrm{crud}} > 1.96$, where $z_{\mathrm{crud}} = |r|/\sqrt{\sigma_K^2 + 1/(n-1)}$) the crud scale must satisfy $\sigma_K \lesssim 0.05$. At $K = 0$ this holds in only two of nine datasets (Stringer, GTEx); after removing ten PCs, only three (Stringer, Kay, HEXACO). In the remaining six datasets, $r = 0.10$ does not clear the threshold at $K = 10$ within the fixed-K model, regardless of n.

C. Mechanism: power-law spectra. Why does removing broad shared variation not shrink the correlation distribution faster? Across all nine datasets, explained variance decays approximately as a power law (Figure 3): the spectrum is roughly linear on log–log axes, meaning shared variation is distributed across many components rather than concentrated in the first few. This contrasts with the common implicit "spike" assumption behind low-rank adjustment: if confounding variance were concentrated in a few dominant components, removing them would collapse the background. Instead, removing the top K PCs eliminates only a modest portion of the dependence structure unless K is large, which is why residual correlations remain broadly distributed even at $K = 10$. Choosing a large K is not a remedy: PCs beyond the leading ones are estimated with

Significance Statement

Observational science routinely interprets small adjusted correlations as evidence for causal effects. Across nine large datasets, most pairs of measured variables are correlated to a non-trivial degree even after standard adjustments, and the typical small adjusted correlation in observational medicine, nutritional epidemiology, common-variant genetics, and brain-wide association studies has a magnitude not distinctive against the same-domain background. We propose the crud-aware percentile rank, which places any reported correlation within the empirical distribution of same-pipeline residuals. The diagnostic is a magnitude check complementary to identification: a small adjusted correlation whose magnitude is in the bulk of the empirical null is not, by itself, evidence of a direct causal relation, even if its classical p-value is tiny.

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The author declares no competing interests.

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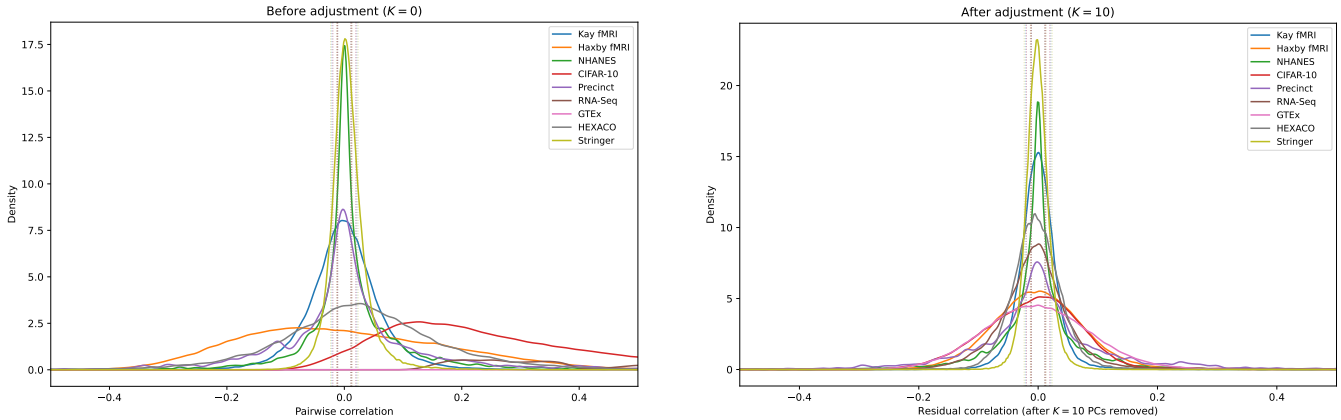

Fig. 2. Broad distributions of correlations persist after generic low-rank adjustment. Each curve shows the density of pairwise correlations across uniformly sampled feature pairs in one dataset (color legend in panel). Left: before adjustment ($K = 0$). Right: after regressing each feature on the top $K = 10$ principal component scores. Dotted vertical lines mark the classical two-sided $p < 0.05$ threshold for each dataset's full n; nearly the entire empirical null exceeds this threshold, so "significant" does not mean "distinctive."

Table 1. Off-diagonal correlation spread (SD) after removing the top K principal components. Entries report the standard deviation of off-diagonal correlations across uniformly sampled feature pairs. The spread generally decreases with K but remains nontrivial in all datasets.

Dataset	$K = 0$	$K = 1$	$K = 10$	$K = 50$
Kay fMRI	0.065	0.055	0.035	0.029
Haxby fMRI	0.175	0.150	0.077	0.038
NHANES	0.115	0.102	0.093	0.117
CIFAR-10	0.191	0.212	0.105	0.064
Precinct	0.166	0.159	0.136	0.293†
RNA-Seq	0.170	0.196	0.071	0.043
GTEx	0.038	0.255	0.090	0.043
HEXACO	0.144	0.125	0.046	0.040
Stringer	0.035	0.031	0.021	0.021

†Precinct has only p=83 features; removing 50 of 83 PCs leaves a near-degenerate residual subspace.

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increasing noise, so removing too many replaces bias with variance (visible for Precinct at $K = 50$). We treat the approximately power-law spectrum as an empirical regularity; mechanism is outside scope. Robustness checks against artifactual explanations are in SI Appendix 7.

D. Bridge lemma: the crud scale and its decay. We formalize the link from the eigenvalue spectrum to the residual correlation distribution. Theorem 2 (SI Appendix 2) shows that under a Haar-like delocalization condition on the eigenvectors of the covariance, the SD of the residual correlation $\rho_{ij}^{(K)}$ across uniformly random pairs is well approximated by a ratio of eigenvalue tail sums:

$$\sigma_K \;:=\; \frac{\sqrt{\sum_{k>K} \lambda_k^2}}{\sum_{k>K} \lambda_k}. \qquad [1]$$

The numerator measures how unevenly the remaining variance is distributed across components; the denominator sets the scale. Important: Table 1 reports the empirical residual-correlation SD measured directly across uniformly sampled feature pairs, not the closed-form prediction Eq. (1). The downstream crud-aware test calibrations therefore rely on direct measurement, not on the spectral formula. The formula's accuracy as a *predictor* of the empirical SD varies across datasets: SI Appendix 3 (Test 3) reports predicted-vs-empirical correlations $r > 0.94$ in seven of nine datasets, with weaker agreement in CIFAR-10 ($r = 0.88$) and RNA-Seq ($r = 0.64$), and Test 4 reports diagonal concentration that holds well in high-p data and degrades in low-p settings (CV up to 1.13 in Precinct, 0.81 in NHANES at $K = 50$). Eigenvector delocalization is also met only approximately (3.8–11.2× the Haar ideal; Test 2). The resulting bias on the spectral formula is conservative for the crud-scale claim where it applies (localization tends to make Eq. (1) overestimate the typical residual correlation; SI Appendix G); where it does not apply quantitatively (low-p, heavy-tailed regimes), the empirical σ_K in Table 1 stands on its own as a direct observation.

If the eigenvalues follow a power law $\lambda_k \approx ck^{-\alpha}$, then σ_K decays slowly: in the especially relevant $\alpha \approx 1$ case, $\sigma_K \approx 1/(\sqrt{K}\log(p/K))$ (Corollary 1, SI Appendix 2). Removing $K = 10$ components can leave a substantial residual crud scale when p is large, formalizing why low-rank adjustment shrinks the correlation distribution only modestly.

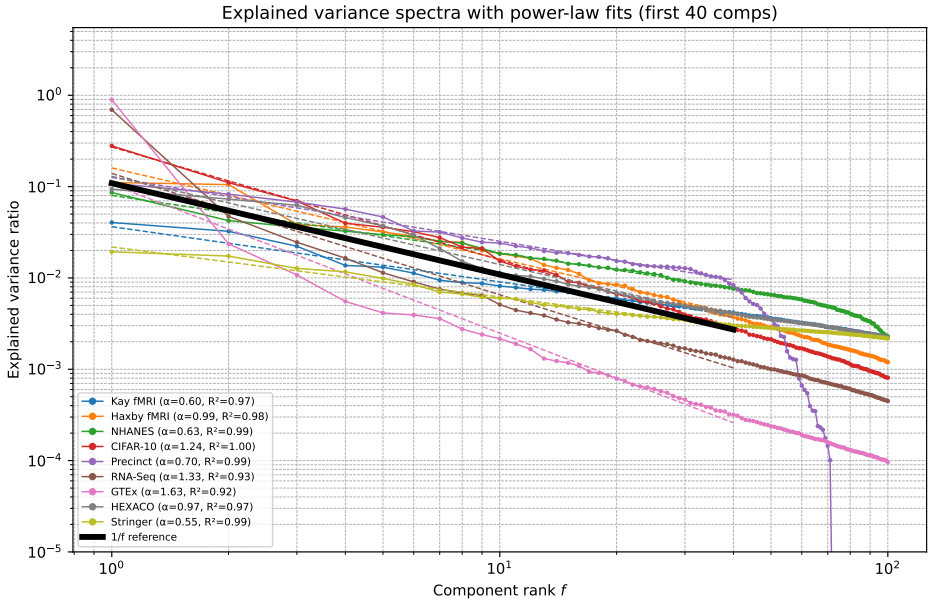

Fig. 3. Explained-variance spectra are approximately power-law across all nine datasets. Curves show the fraction of variance explained by each principal component on log–log axes; fitted exponents $\alpha \in [0.55, 1.33]$ (shown in panel legend along with R^2 of the power-law fit). A straight line on these axes means a power law $k^{-\alpha}$; when α is near 1, variance is spread broadly across components, so removing the first few PCs leaves most of the shared dependence intact. The "$1/f$ reference" line is shown for comparison.

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σ_K as a pipeline functional. σ_K is a population functional of the joint $(\Sigma, K, \text{preprocessing pipeline})$, not of the data alone. Estimating it requires a fixed pipeline (z-scoring policy, missing-data handling, PCA implementation, sub-sampling); changing any of these changes σ_K. Cross-paper or cross-domain σ_K comparisons therefore require pipeline matching, not just dataset matching, and are best interpreted within a single fixed pipeline (SI Appendix H).

Asymptotic regime. Different theorems hold different quantities fixed: Theorem 2 is a $p \to \infty$ statement at fixed or slowly growing K; Theorem 3 (decision-theoretic) holds σ_K fixed; Theorem 1 (parametric crud-aware test) holds p fixed and varies n. The "regardless of sample size" claim in the abstract refers specifically to the fixed-(p, K) regime of Theorem 1: the parametric variance does not vanish with n when σ_K^2 is the dominant term. We do not claim joint uniformity across (n, p, K); statements at different limits should be read independently.

E. Decision-theoretic limit. A complementary question: how well can magnitude alone classify a pair as causally linked versus correlated by background dependence? Theorem 3 (SI Appendix 2) gives the Bayes-error benchmark for this magnitude-only problem. Under a stylized Gaussian mixture in which a direct causal edge induces a mean shift μ in the adjusted association statistic T_{ij}, the minimum achievable misclassification error for any rule using only T_{ij} is $\Phi(-|\mu|/(2\sigma_K))$. When $|\mu| \lesssim \sigma_K$, this rate is a constant bounded away from zero. Two clarifications matter. First, this is the irreducibility of magnitude-only classification under this mixture, not a claim about what inference can recover about ρ_{ij} itself: an analyst with a sample correlation $\hat{\rho}_{ij}$ estimates ρ_{ij} ever more precisely as $n \to \infty$. The benchmark reflects overlap between the two classes in the mixture, not an obstruction to estimation. Second, whether the model-level σ_K matches the conceptually natural "background dependence" in a given domain depends on whether the chosen adjustment captures that domain's confounding structure (SI Appendix 4). Sparse causal structure ($\pi \ll 1$) further compounds the magnitude-only limitation by collapsing positive predictive value (SI Appendix F). Heavy-tailed residuals, additional variance structure, and selected target pairs are discussed in SI Appendix 2 and SI Appendix 4.

F. Crud-aware testing. The empirical results above imply a minimum reporting standard for small-effect observational claims: compare any adjusted association to the same-pipeline empirical null—the distribution of adjusted associations over uniformly sampled feature pairs from the same dataset, computed under identical preprocessing and adjustment. The resulting crud-aware p-value is operationally a percentile rank: the fraction of same-pipeline reference pairs whose absolute adjusted association exceeds the target's. Despite the p-value notation, this is a magnitude-rank diagnostic rather than a frequentist test of a sharp null at $\rho = 0$; SI Appendix 2 clarifies the status. Associations below a chosen high percentile of this null (conventionally the 95th) should be reported as not distinctive against the domain background; the cutoff is conventional, not principled. The test is a magnitude diagnostic: failing to clear it means the magnitude alone cannot be cleanly distinguished from typical residual dependence, not that the relation is causally unimportant (mediation, suppression, and confounder-driven attenuation can produce small adjusted correlations from causally important relations); clearing it is necessary, not sufficient, for a causal interpretation. SI Section 4 gives details, the handling of selected target pairs (which violate exchangeability), stratified comparisons, and the relation to multiple-testing corrections.

G. A parametric crud-aware z-test. Many studies measure only a few variables on a small sample. Such a study can be understood as a small submatrix of the much larger data matrix one could have collected; the crud factor operates at the domain level. Under a Gaussian approximation (Theorem 1, SI Appendix 2), the total variance of r for a random pair from the domain is the sum of crud and sampling contributions, $\mathrm{Var}(r) \approx \sigma_K^2 + 1/(n-1)$, where σ_K is the population residual-correlation SD defined in Eq. 1. This yields the test statistic

$$z_{\mathrm{crud}} = \frac{|r|}{\sqrt{\sigma_K^2 + 1/(n-1)}}, \qquad [2]$$

with two-sided p-value $p_{\mathrm{crud}} = 2\,\Phi(-z_{\mathrm{crud}})$. The classical correlation test is the special case $\sigma_K = 0$. As n grows, $1/(n-1)$ shrinks but σ_K^2 does not, so for the purpose of distinguishing small associations from background dependence there are diminishing returns past $n \approx 1/\sigma_K^2$ (SI Appendix B). The Gaussian formula is broadly calibrated at the conventional 5% threshold and loses calibration for $p_{\mathrm{crud}} < 0.01$ in heavy-tailed domains; for stronger claims use the empirical null (SI Appendix 2, Table S1; SI Appendix C). A step-by-step reporting workflow, an online calculator, and guidance on choosing σ_K for an applied study are in SI Appendix D.

H. Worked example: folate–mortality in T2D. (author?) (9) report a nominally significant association between folate and mortality in adults with type-2 diabetes ($r \approx 0.02$, $p_{\mathrm{classical}} \approx 0.04$, $n \approx 8{,}000$). To compare on a same-pipeline reference, we computed σ_K on the T2D-restricted NHANES sub-cohort directly (HbA1c $\geq 6.5\%$, $n = 1{,}912$, $p = 154$): $\sigma_K = 0.104$ at $K = 10$, slightly larger than the full-cohort value ($\sigma_K = 0.093$). The crud-aware z-test gives $z_{\mathrm{crud}} = 0.19$, $p_{\mathrm{crud}} = 0.85$: the effect's magnitude is typical of randomly chosen pairs in the T2D sub-cohort after the same adjustment, and the verdict is essentially unchanged from the full-cohort comparison ($p_{\mathrm{crud}} = 0.83$). By contrast, within-facet HEXACO item pairs (designed to load on the same trait) sit far in the upper tail of the empirical null ($p_{\mathrm{crud}} = 0.03$ for the median pair). SI Appendix I gives both examples in full and reports two further cases (NHANES biomarkers, GTEx cis-eQTL effects).

3. Discussion

When reported adjusted associations are comparable to the crud scale, causal interpretation is fragile without additional design leverage, even if classical p-values are tiny. The crud-aware test provides a minimal magnitude standard for deciding when a small association is worth interpreting. As positive controls, smoking and lung cancer (RR≈ 7, converted to a Pearson correlation via Eq. Eq. (13) of SI Appendix C at representative population prevalences gives $r \approx 0.28$, or $\approx 3\sigma_K$ against NHANES at $K = 10$) and within-facet HEXACO personality items (best pair >99.99th percentile) clear the threshold; SI Appendix A gives the structural-ceiling caveat for binary correlations and the case-control-vs-cohort distinction.

Scope and leverage. Prior knowledge or design-based leverage—randomization, discontinuities, valid instruments, mechanistic priors—effectively isolates variables from background dependence and is orthogonal to generic correlation structure. Our results target association-only workflows that interpret small adjusted pairwise correlations as causal evidence in the absence of additional leverage. The crud-aware test is a magnitude diagnostic complementary to identification (SI Appendix C); it asks "is the magnitude distinctive against same-pipeline residual dependence," not "is the effect identified." The test is most useful when identification is uncertain, where it adds a magnitude floor on top of an unverified identification argument; it is least useful when identification is clean (e.g., a well-designed RCT), where magnitude alone is already meaningful relative to a sharp null.

Empirical comparison to standard alternatives. On a synthetic data-generating process with latent-factor confounding and a small set of embedded direct-edge pairs (SI Appendix G), averaged over 50 seeds, the crud-aware diagnostic at $K = 10$ recovers 98% of direct edges at the conventional 95th-percentile rank with precision ≈ 0.05 (30× better than classical), and recovers 90% at the 99th percentile with F1 ≈ 0.36. This dominates classical $|r| > 1.96/\sqrt{n-1}$, BH-FDR (theoretical and dependent-BY variants), and Efron's empirical-null + BH (10) on every metric: BY-FDR and Efron+BH improve modestly over classical BH but still fail to control nominal FDR because they only recalibrate the marginal distribution of test statistics, not the structural confounding that PC adjustment removes.

Real small effects supported by RCT/mechanism would also fail this test. Several real, clinically important interventions land at small RR's that the magnitude-only test would not clear: aspirin in primary cardiovascular prevention (pooled RR ≈ 0.94 for CV mortality), low-dose statins (RR ≈ 0.78 for major CV events), ACE inhibitors in non-diabetic CKD progression. This is not a failure of the test—it is the correct diagnosis: *the magnitude alone is not what makes these effects compelling.* What carries each case is RCT replication, mechanism, dose-response, removal effect, and biological plausibility—design leverage on top of small magnitudes. The test correctly diagnoses small-RR clinical effects as not magnitude-distinctive, which is consistent with their being supported by other lines of evidence rather than by raw correlation.

The mirror-image objection—"my mechanism privileges this pair, so the random-pair null is the wrong reference"—is addressed in SI Appendix E: in practice, "I had a mechanism" is often a post-hoc rationalization of one story among many, and the magnitude distribution of mechanism-selected pairs in the literature is much closer to the random-pair distribution than introspection suggests. σ_K is itself a pipeline-relative quantity (SI Appendix D); cross-paper comparisons require pipeline matching.

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What to do when an effect does not clear the test. Failing the test is not a verdict that the relation is causally unimportant; it is a verdict that the magnitude alone cannot carry the case. Constructive next steps are: (i) seek design leverage—randomization, instruments, regression discontinuities, negative controls; (ii) pre-register the hypothesis and replicate in an independent dataset; (iii) accept that converging evidence (mechanism, dose–response, removal effect, prior literature) can carry a causal claim even when magnitude alone does not, as in the smoking–cancer case. For fields where design leverage is structurally unavailable—nutritional, environmental, and lifestyle epidemiology—SI Appendix F gives a magnitude-as-descriptor fallback.

Limitations. All results concern linear dependence (correlations, linear PC adjustment, covariance spectra); σ_K is a lower bound on background dependence in domains with substantial nonlinear structure. The quantitative σ_K varies across domains and preprocessing; the decision-theoretic limit is per-pair, equal-variance, Gaussian, and fixed-K. Three foundational issues are out of scope here and remain open: an analytic correction for selected target pairs (we recommend pre-registration and stratified references but provide no closed-form fix); whether the empirical null reflects latent dependence or unmodeled adjustment (left open by the test, which is calibrated for the magnitude question regardless of which interpretation prevails); and broader robustness of the impossibility theorem beyond the stylized Gaussian-mixture model.

Materials and Methods

Datasets.. Nine large public datasets spanning population health, neuroscience, genomics, psychology, political science, and vision (Section A; SI Appendix 1 gives provenance and dimensions): NHANES, precinct voting + Census, GTEx v8, Allen Institute mouse RNA-seq, HEXACO, CIFAR-10, Kay fMRI, Haxby fMRI, and Stringer mouse V1.

Preprocessing.. Each feature is z-scored (mean zero, unit variance). Sample sizes used in analysis are capped at $n \approx 10{,}000$ per dataset for computational tractability where the raw n is larger.

Adjustment.. The deliberately generic adjustment is regression on the top K principal component scores, with $K = 10$ as the headline choice and sensitivity reported across $K \in \{0, 1, 10, 50\}$ (Table 1). PCA is computed via randomized SVD with up to 300 components. Alternative adjustments (Lasso) are reported in SI Appendix D.

Empirical null and crud scale.. For each dataset and each K, we compute residual correlations across uniformly sampled feature pairs (or, where $\binom{p}{2} \leq 5 \times 10^5$, all pairs). The crud scale σ_K is the SD of these residual correlations. Eigenvalue spectra are fit by power-law on the first 40 components against an exponential alternative via AIC.

Statistical analysis.. The crud-aware z-test compares an observed correlation r to σ_K via Equation Eq. (2). The full empirical null replaces the Gaussian formula with the empirical fraction of reference pairs whose absolute adjusted association exceeds the target's. Calibration of the parametric formula against the empirical null in heavy-tailed datasets is reported in SI Appendix 2, Table S1. A reference catalog mapping common application domains to a recommended σ_K is in SI Appendix H.

Code and data.. All analysis is in the Python package at https://github.com/koerding/crud. An interactive demonstration and a standalone crud-aware test calculator are at https://koerding.github.io/crud/ and https://koerding.github.io/crud/test.html.

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